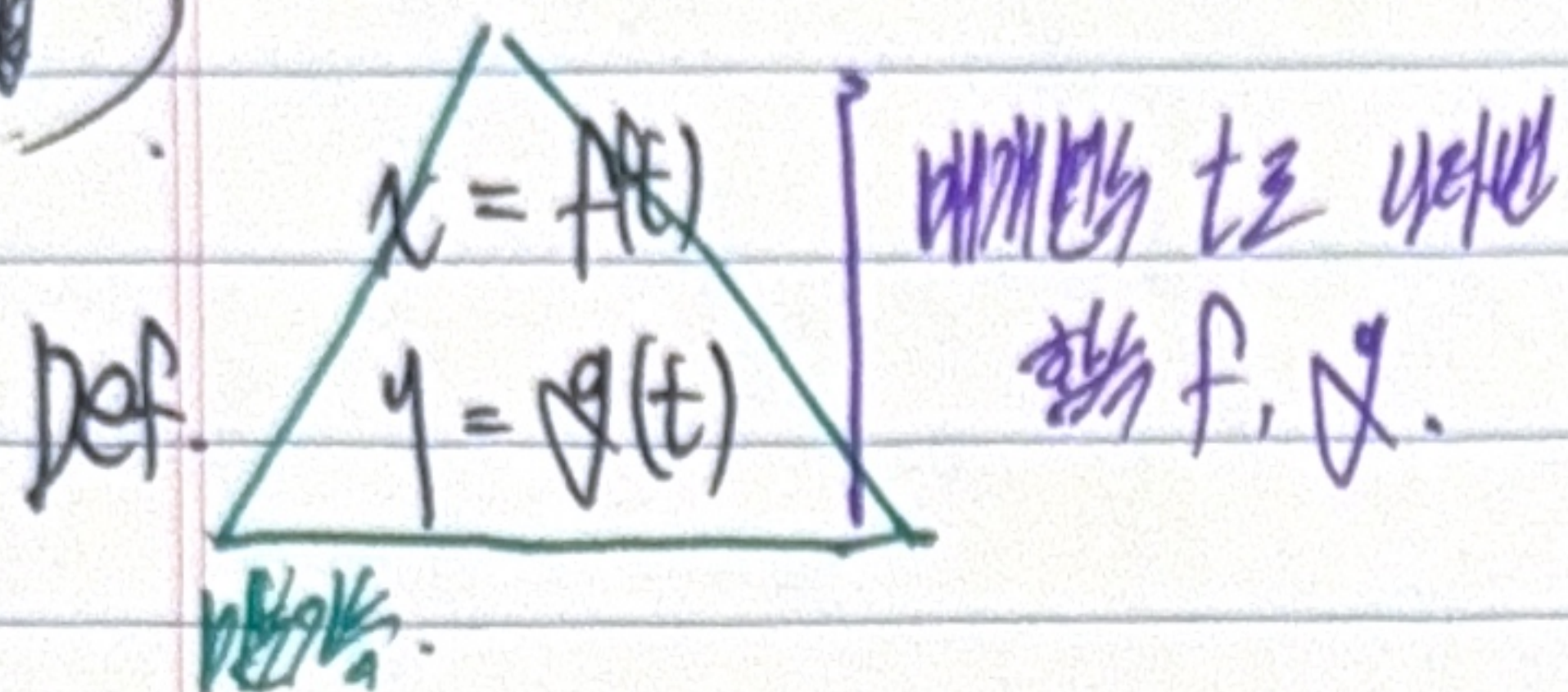


chain rule



$$\Rightarrow \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{g'(t)}{f'(t)} \neq 0.$$

chain rule

• chain rule is used to find the derivative of a function of a function.

• chain rule $\Rightarrow$ find the derivative of a function of a function.

Ex)

$$x = 3t - 2.$$

$$\frac{dx}{dt} = 3.$$

Ex2) $1 + 2t^2 = y$

$$\frac{dy}{dt} = 4t.$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4t}{3}$$

$$\therefore \frac{dy}{dx} = \frac{4}{3} t \quad \square$$

chain rule is used to find the derivative of a function of a function.

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

chain rule is used to find the derivative of a function of a function.

$\frac{dy}{dx}$ is the derivative of y with respect to x.